

DFD Field Report System

Mobile-Compatible Web Application

Version 1.0

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1. System Overview

The DFD Field Report system guides a mobile user through a clear Level 1 data flow: the actor submits validated input, data is stored relationally, server processes retrieve records with role-based rules, and a structured JSON dashboard is rendered in the browser with optional PDF export.

2. Hardware Requirements

- Mobile device: Android 8.0+ or iOS 13+
- Minimum RAM: 2 GB
- Internet: 3G / 4G / Wi-Fi
- Minimum resolution: 360×640
- Free storage: at least 50 MB for browser cache and downloads

3. Software Requirements

- Modern mobile browser: Safari (iOS), Chrome (Android)
- Backend: Node.js 18+ with Express (REST + JWT)
- Database: MySQL 8+
- Client: React (Vite build) with Tailwind CSS

4. Installation / Access Guide

- 1) Install Node.js and MySQL on a workstation or server.
- 2) Import database/schema.sql.
- 3) Configure server/.env from server/.env.example (DB + JWT secret).
- 4) Run npm install && npm run seed inside /server, then npm run dev.
- 5) Run npm install && npm run dev inside /client.
- 6) From your phone, browse to <http://<computer-ip>:5173> on the same LAN (or HTTPS host in production).

5. User Roles

Admin: can see all field reports and all activity logs tied to the datastore.

Regular user: can create and manage their own reports; sees only their audit log entries.

6. Step-by-Step Usage Guide

Step 1 — Open the app URL in your mobile browser.

Step 2 — Register or sign in with email + password (JWT session).

Step 3 — On Home, fill the field report form (title, category, description, priority, status).

[Screenshot placeholder: input form on iPhone SE viewport]

Step 4 — Tap Submit; the server validates and stores the record, writing an activity_logs row.

Step 5 — Open Results to view cards + table sorting (business rules).

Step 6 — Tap Export PDF to generate a handset-friendly report.

7. DFD Level 1 — Educational grading reference

Structured from the printed coursework figure (Educational Management & Grading):
Instructor, Student, and Administrator as external entities;
processes P3–P19; data stores D1–D16; outputs on the diagram right side.
Full scan: docs/educational-dfd-level1-reference.png

7.A Processes P3–P19

P3 Login · P4 Add Teaching Load · P5 Teaching Load Details · P6 Grading Composition ·
P7 Base Grade · P8 View Class Record · P9 Add Activity · P10 Record Score ·
P11 Calculate Grade · P12 Record Attendance · P13 Enter Class Record ·
P14 User Account Mgmt · P15 System Logs · P16 Teaching Load Mgmt ·
P17 Subjects Mgmt · P18 Course Mgmt · P19 Roles Mgmt

7.B Data stores D1–D16

D1 Users · D2 Roles · D3 Teachers · D4 Semester · D5 Students · D6 Subjects ·
D7 Teaching Loads · D8 TL Details · D9 Grade Category · D10 Grading Composition ·
D11 Grade Base · D12 Grading · D13 Term · D14 Grading Detail ·
D15 Enrollments · D16 Courses

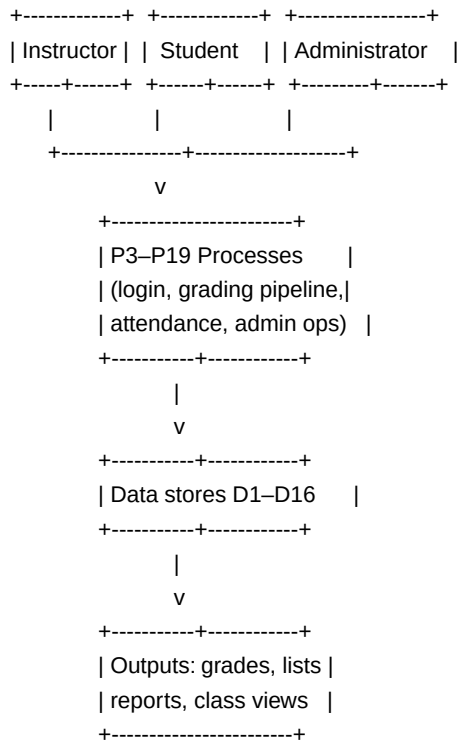
7.C Key inputs (actors !' processes)

Instructor: credentials!'P3; weights!'P6; base grade!'P7; activity!'P9; student score!'P10.
Student: scan ID!'P12; class QR!'P13.
Administrator: users!'P14; teaching load!'P16; subjects!'P17; courses!'P18; roles!'P19.

7.D Outputs (views / reports)

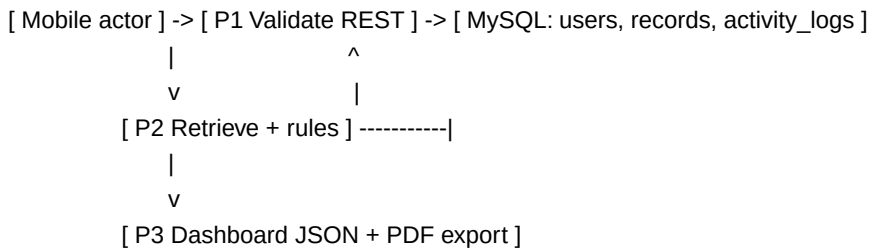
View User Details · View Teaching Loads / Details · View Class Record · Recorded Score ·
Student grade per Term & Semester · Enrolled Subject · Student/Teacher Lists · Report ·
Teaching Load List · Subjects List · Course List

7.E Reference flow — ASCII overview



Central logic: P11 Calculate Grade uses D9–D14 with recorded scores (P10) and composition/base (P6/P7). Administration uses P14–P19 to maintain D1–D8,D16.

7.F Simplified running prototype (this codebase)



8. Troubleshooting

- Cannot login: verify API health (/api/health) and MySQL credentials in server/.env.
- CORS failures: CLIENT_URL must match the origin you type in the mobile browser.
- Empty dashboard after submit: refresh; ensure token still valid (JWT expiry).
- Blank PDF export: populate data on Results screen first.

9. Contact / Support

Support Email: support@dfd-field-report.local

Knowledge base: README.md inside /dfd-mobile-web